

# What we're doing

Every MarinerFold eval so far scores contact prediction. contacts-and-crops-v1 (#130) encodes real 3D positions — Pass-1 coarse 10 Å boxes plus Pass-2 fine 0.1 Å crops — so a model trained on it should be able to emit an actual structure.

This experiment asks whether it can, scored the way structure predictors are normally scored: CA-RMSD, all-atom RMSD, IDDT, IDDT-CA and TM-score against the 554-protein eval set from #74/#78.

# Two components, two gates

Component 1 — inference (document to coordinates) — is plans-first by design. Six approaches are written up in PLANS.md. Plan F is now the plan of record: neighbour-conditioned iterative refinement. Not yet implemented.

Component 2 — scoring — is done. It takes a directory of predicted coordinate files in a documented one-PDB-per-protein contract and needs to know nothing about how they were produced. Ground truth, model predictions and baselines all speak the same contract.

# Ground truth: 554/554, and one library bug

Full-atom ground truth for all 554 records: 777,459 heavy atoms, zero failures, minimum sequence-alignment identity 0.95, lengths 30 to 761.

Building it surfaced a real bug in the shared library. The coordinate parse layer joins the pyconfind residue list to the gemmi atom walk on (chain, residue number), and the two libraries spell a blank author chain id differently — gemmi "", pyconfind "\_". Structures with no chain id joined nothing and produced a document with an empty coordinate section, silently. AFDB is always chain A so the training corpus never hit it; 19 of the 554 eval proteins did. Fixed, with the parse layer now raising instead of emitting a coordinate-free document.

# Partial predictions are the normal case

About 96% of contacts-and-crops-v1 documents are truncated by construction, so the harness reports two families of metric and keeps them apart.

Coverage-penalized (IDDT, IDDT-CA, TM-score) take their denominators from the ground truth, so an atom the model never placed costs score. These are the model-comparison numbers.

Covered-only (RMSD, and the "covered" IDDT variants) are computed over the atoms that were placed. A predictor that emits three atoms perfectly scores 0.0 Å RMSD, so these are meaningless without the coverage columns beside them.

A record with no prediction file is scored as a total miss, not skipped.

# The headline so far: the format's ceiling

Before any model number means anything, we measured what a perfect model could score. Degrade the ground truth to each of the format's resolution tiers and run it through the same harness.

The 0.1 Å digit vocabulary costs nothing: IDDT 1.000, TM 1.000. But every atom at its correct 10 Å box centre scores only IDDT 0.32 / TM 0.51, and one realistic document — 65% of atoms boxed, 25% refined — tops out at IDDT 0.17 / TM 0.41.

IDDT falls as coverage squared (a contact needs both its atoms); TM-score falls linearly (a residue needs only itself).

# What that implies for inference

An inference plan that samples one document per protein is competing for IDDT 0.17. A result from such a plan cannot distinguish "the model cannot fold" from "the format did not get a chance", which is the central argument in PLANS.md for spending inference compute to raise the refined fraction.

Inference cost is not the constraint: a full 8192-token document for all 554 proteins is about three GPU-minutes on one H100.

The Pass-2 refined fraction is the entire ballgame. Going from 15% to 50% refined moves the achievable IDDT from 0.36 to 0.53 and TM from 0.58 to 0.75. That is a format finding as much as an inference one.

# The plan of record: a scanning flashlight

Plan F. Generate Pass 1 once, which fixes the frame. Then sweep a spatially coherent path over the occupied voxels: for each one, build a prompt ending in its crop header, draw K sampled crop bodies, and fold them into a running precision-weighted per-atom estimate. Repeat the sweep, because a voxel's neighbours have moved since it was last visited, until the coordinates stop moving.

Two things make this fit the format rather than fight it. Each crop is conditioned on its already-refined neighbours, and Pass-2's own box selection is 45% frontier and 10% re-show, so a local scan with revisits is the training-time crop distribution, not a departure from it. And the prompt — sequence, full Pass 1, up to about twenty prior crops, one new header — is exactly the shape of a real training document, because the fine reserve holds about twenty crops.

Sampling, not greedy: averaging K draws recovers the same estimate and hands back a per-atom variance for free, which is what the B-factor column and the precision weighting both want. Two independent temperatures, one for coordinate tokens and one for structural choices.

Cost: about 1.5 to 2 H100-hours for all 554 proteins, provided the synthesized Pass-1 section is held byte-identical within a sweep so the prefix cache hits. That one decision is worth 10x.

# Status

Harness built, tested (34 tests) and validated against a measured ceiling. Inference approach agreed: Plan F. Remaining work is E2 and E3 as gates, A and C as controls, then F itself, then scoring both #137/#155 checkpoints — which are staged and ready.



## ceiling.png

Model-free ceiling on the 554-protein eval set: ground truth degraded to each contacts-and-crops-v1 resolution tier, scored by the same harness. Left: IDDT falls as  $\text{coverage}^2$ , TM-score linearly. Right: the Pass-2 refined fraction is the whole ballgame; a single document reaches IDDT 0.17 / TM 0.41.

